

ME 2261 - Embedded Systems Log BookTask 01

Find the contents of effective registers and the contents of the \$65 memory locations.

LDI r17, \$33

LDI r18, \$21

Inc r17

Inc r17

Dec r18

Add R17, R18

STS \$65, R17

Steps	R17	R18	0x0065
1	\$33		\$00
2		\$21	\$00
3	\$34		\$00
4	\$35		\$00
5		\$20	\$00
6	\$55	\$20	\$00
7	\$55		\$55

1	.INCLUDE "M32DEF.INC"	Memory 1
2	.org \$00	Memory: data REG
3		
4	begin:	
5	LDI r17, \$33	data 0x0065 55
6	LDI r18, \$21	data 0x007A 00
7	Inc r17	data 0x008F 00
8	Inc r17	data 0x00A4 00
9	Dec r18	data 0x00B9 00
10	Add R17, R18	data 0x00CE 00
11	STS \$65, R17	data 0x00E3 00
12		data 0x00F8 00
13		data 0x010D 00
14	end:NOP	data 0x0122 00
		data 0x0137 00

Task 02

Find the contents of effective registers.

Steps	R17	R18	R19
1	\$9F		
2		\$61	
3			\$00
4	\$00	\$61	
5	\$01		

1	.INCLUDE "M32DEF.INC"		
2	.org \$00		
3			
4	begin:		
5	LDI r17, \$9F		
6	LDI r18, \$61		
7	CLR R19		
8	Add R17, R18		
9	ROL R17		
10		R17	0x01
11		R18	0x61
12	end:NOP	R19	0x00

Task 03

Steps	R0	R1	R17	R18
1			\$90	
2				\$30
3			\$09	
4				\$03
5	\$1B	\$00	\$09	\$03
6	\$1B		\$1B	
7		\$00		\$00

```
1  .INCLUDE "M32DEF.INC"
2  .org $00
3
4  begin:
5  LDI r17, $90
6  LDI r18, $30
7  Swap R17
8  Swap R18
9  mul R17, R18
10 mov R17, R0
11 mov R18, R1
12
13                                R00          0x1B    R17          0x1B
14  end:NOP                      R01          0x00    R18          0x00
```

Task 04

Subtract two 8 bit numbers without using SUB, SBC and SUBI function

```
1  .INCLUDE "M32DEF.INC"
2  .org $00
3
4  begin:
5  LDI r17, $8
6  LDI r18, $5
7  NEG R18
8  add r17,r18
9
10
11  end:NOP
```

Task 05

Find the contents of effective registers

Steps	R17	R18
1	\$90	
2		\$30
3	\$C8	
4		\$18
5	\$37	
6		\$E8

```
1  .INCLUDE "M32DEF.INC"
2  .org $00
3
4  begin:
5  LDI r17, $90
6  LDI r18, $30
7  asr r17
8  asr r18
9  com r17
10 neg r18
11                                     R17      0x37
12 end:NOP                             R18      0xE8
```

Task 06

Toggle selected bits in the output PORTB using EOR

Initialize the register R18 with 0b10101010.

- ✓ Send R18 register value to PORTB
- ✓ Keep a 5-microsecond delay (5 clock cycles delay)
- ✓ Toggle bits 0 to 3 using appropriate bitmask (Pattern of bits) and EOR.
- ✓ Store the results in R19
- ✓ Send R19 register value to PORTB keep a 5 microsecond delay.

I/O PORTB 0x38 0xAA 

I/O PORTB 0x38 0xA5 

```
1  .INCLUDE "M32DEF.INC"
2  .org $00
3
4  begin:
5  LDI R18, 0b10101010
6  OUT PORTB, R18
7  NOP
8  NOP
9  NOP
10 NOP
11 NOP
12 LDI R19, 0b00001111
13 EOR R19, R18
14 OUT PORTB, R19
15 NOP
16 NOP
17 NOP
18 NOP
19 NOP
20 end:NOP
```

Task 07

Multiply two 8-bit numbers and store the results in the Z memory location.

A. Initialize values: R25 = 0x0A and R26 = 0x04

1. What is the result in decimal and hexadecimal?

Hexadecimal = 0x0028

Decimal = 40

2. What happens to the Carry flag?

Carry flag is always in 0 state

B. Initial values: R25 = 0xA0 and R26 = 0x44.

1. What is the result in decimal and hexadecimal?

Hexadecimal = 0x2A80

Decimal = 10880

2. What happens to the Carry flag?

Carry flag is always in 0 state

1	.INCLUDE "M32DEF.INC"
2	.org \$00
3	
4	begin:
5	LDI R25, \$A0
6	LDI R26, \$44
7	MUL R25, R26
8	MOV R31, R1
9	MOV R30, R0
10	end:NOP

Processor	
Name	Value
Program Counter	0x00000005
Stack Pointer	0x0000
X Register	0x0044
Y Register	0x0000
Z Register	0x2A80
Status Register	I T H S V N Z C